

Feedforward Neural Networks in Depth

[coursera-platform](#)

[jonaslalin](#) 1 February 14, 2022, 9:54am

Forward propagation, backward propagation, various activation functions, various cost functions, vectorization... 🤖

Would you like to read my derivation of the mathematics behind feedforward neural networks?

[Feedforward Neural Networks in Depth, Part 1: Forward and Backward Propagations](#)

This post is the first of a three-part series in which we set out to derive the mathematics behind feedforward neural networks. They have

[Feedforward Neural Networks in Depth, Part 2: Activation Functions](#)

This is the second post of a three-part series in which we derive the mathematics behind feedforward neural networks. We worked our way through forward and backward propagations in the first post, but if you remember, we only mentioned activation...

[Feedforward Neural Networks in Depth, Part 3: Cost Functions](#)

This post is the last of a three-part series in which we set out to derive the mathematics behind feedforward neural networks. In short, we covered forward and backward propagations in the first post, and we worked on activation functions in the...

549 Likes

[How does a Deep Neural Network work?](#)

[Optional Reading in Week 4](#)

[armanbm2013](#) 3 March 29, 2022, 8:26am

jonaslalin:

[Feedforward Neural Networks in Depth, Part 1: Forward and Backward Propagations | I, Deep Learning](#)

excellent in depth explanation about Forward and backward propagation
thanks for sharing it with us

5 Likes

[Steven Kao](#) 4 April 2, 2022, 2:31am

Great resource thank you for putting it out Jonas.

One question I have with the Part 1 article - the definition of the activation seems a bit puzzling to be. Now it is defined as

$$a_{j,i}^{[l]} = g_j^{[l]}(z_{1,i}^{[l]}, \dots, z_{j,i}^{[l]}, \dots, z_{n^{[l]},i}^{[l]}).$$

But I thought the activation function (g_j) is a function that takes in scalar value and also outputs scalar value. So I thought the equation should be

$$a_{j,i}^{[l]} = g_j^{[l]}(z_{j,i}^{[l]})$$

This is at least my understanding from the current course 1. Would appreciate if anyone points out what I might have misunderstood... Thanks a lot again 🙏

34 Likes

[jonaslalin](#) 5 April 2, 2022, 6:51am

Great question. If you check out the second article in the series, you will find that I also derive the math for the softmax activation function. Since it uses an average, the values of all other nodes in the current layer are used to compute the value for one node. Hence, I need a more general formulation of the activation function than used in the lectures, where the derivation is left as an exercise.

19 Likes

[Steven Kao](#) 6 April 2, 2022, 2:58pm

Thank you for the clarification! So I suppose this formula will start to make more sense once I start getting into the second course where we have multi-label classification

4 Likes

[jamesturner246](#) 8 April 17, 2022, 8:53am

Hi [@jonaslalin](#) .

I note that you do not use a $1/m$ term in (13) and (14) of post one - i.e. your gradients are summed, rather than being averaged as in the lectures. Is this intentional?

In other words, should one average or sum the gradients of dw and db over all training examples?

Thanks.

2 Likes

[jonaslalin](#) 9 April 18, 2022, 11:55am

Yes this is intentional. In the lectures you compute dL/dW , dL/db . I never introduce a “loss” function, only the cost function J .

3 Likes

[ljlz163](#) 11 May 8, 2022, 3:15pm

23:06 Sun May 8 jonaslalin.com

$$u_i = g_i(x_1, \dots, x_j, \dots, x_n), \quad (10)$$

$$y_k = f_k(u_1, \dots, u_i, \dots, u_m). \quad (11)$$

Note that x_j may affect $u_1, \dots, u_i, \dots, u_m$, and y_k may depend on $u_1, \dots, u_i, \dots, u_m$; thus,

$$\frac{\partial y_k}{\partial x_j} = \sum_i \frac{\partial y_k}{\partial u_i} \frac{\partial u_i}{\partial x_j}. \quad (12)$$

Great! If we ever get stuck trying to compute or understand some partial derivative, we can always go back to (10), (11), and (12). Hopefully, these equations will provide the clues necessary to move forward. However, be extra careful not to confuse the notation used for the chain rule example with the notation we use elsewhere in this series. The overlap is unintentional.

Now, let us concentrate on the task at hand:

$$\frac{\partial J}{\partial w_{j,k}^{[l]}} = \sum_i \frac{\partial J}{\partial z_{j,i}^{[l]}} \frac{\partial z_{j,i}^{[l]}}{\partial w_{j,k}^{[l]}} = \sum_i \frac{\partial J}{\partial z_{j,i}^{[l]}} a_{k,i}^{[l-1]}, \quad (13)$$

why?

$$\frac{\partial J}{\partial b_j^{[l]}} = \sum_i \frac{\partial J}{\partial z_{j,i}^{[l]}} \frac{\partial z_{j,i}^{[l]}}{\partial b_j^{[l]}} = \sum_i \frac{\partial J}{\partial z_{j,i}^{[l]}}. \quad (14)$$

Vectorization results in

$$\begin{bmatrix} \frac{\partial J}{\partial w_{1,1}^{[l]}} & \dots & \frac{\partial J}{\partial w_{1,k}^{[l]}} & \dots & \frac{\partial J}{\partial w_{1,n^{[l-1]}}^{[l]}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial J}{\partial w_{j,1}^{[l]}} & \dots & \frac{\partial J}{\partial w_{j,k}^{[l]}} & \dots & \frac{\partial J}{\partial w_{j,n^{[l-1]}}^{[l]}} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \frac{\partial J}{\partial w_{n^{[l]},1}^{[l]}} & \dots & \frac{\partial J}{\partial w_{n^{[l]},k}^{[l]}} & \dots & \frac{\partial J}{\partial w_{n^{[l]},n^{[l-1]}}^{[l]}} \end{bmatrix}$$

Why this happen? Thanks in advance.

2 Likes

[jonaslalin](#) 12 May 8, 2022, 4:27pm

I am very busy at the moment, but I will respond to your question as soon as I can. I believe more learners have the same question. It is true that I need to explain how I reach 13 and 14 in more detail.

1 Like

[Miles Zhou](#) 13 May 11, 2022, 4:24am

keep in mind that $Z[l] = W[l]A[l-1] + b[l]$. since we are trying to find dZ/dW , then we treat everything except dW as constant. thus $dZ/dW = A[l-1]$

6 Likes

[TaeefNajib](#) 14 May 11, 2022, 9:52pm

I don't know if I could help, but here's my answer:

As we know, for the l-th layer, $z = w[l]a[l-1] + b[l]$

So, $dz[l] / dw[l] = d(w[l]a[l-1] + b[l]) / dw[l] = a[l-1] + 0 = a[l-1]$

4 Likes

[ljz163](#) 15 May 29, 2022, 4:51am

That really helps me. Thanks.

1 Like

[Thunderstroke](#) 16 June 25, 2022, 4:08pm

Amazing notes.

Thanks for your kindly explanation.

[xujiali](#) 17 July 6, 2022, 4:31pm

By applying the chain rule, we get

$$\begin{aligned}
 \frac{\partial a_{j,i}^{[l]}}{\partial u_5} &= 1, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_4} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_5} \frac{\partial u_5}{\partial u_4} = u_1 = \exp(z_{j,i}^{[l]}), \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_3} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_4} \frac{\partial u_4}{\partial u_3} = -u_1 \frac{1}{u_3^2} = -\frac{\exp(z_{j,i}^{[l]})}{(\sum_p \exp(z_{p,i}^{[l]}))^2}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_1} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_3} \frac{\partial u_3}{\partial u_1} + \frac{\partial a_{j,i}^{[l]}}{\partial u_5} \frac{\partial u_5}{\partial u_1} \\
 &= -u_1 \frac{1}{u_3^2} + u_4 \\
 &= -\frac{\exp(z_{j,i}^{[l]})}{(\sum_p \exp(z_{p,i}^{[l]}))^2} + \frac{1}{\sum_p \exp(z_{p,i}^{[l]})}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_{-1}} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_1} \frac{\partial u_1}{\partial u_{-1}} \\
 &= \left(-u_1 \frac{1}{u_3^2} + u_4\right) \exp(u_{-1}) \\
 &= -\frac{\exp(z_{j,i}^{[l]})^2}{(\sum_p \exp(z_{p,i}^{[l]}))^2} + \frac{\exp(z_{j,i}^{[l]})}{\sum_p \exp(z_{p,i}^{[l]})}.
 \end{aligned}$$

Hello, I am confused how did you get the formula with red line in my snapshot of Part2, Softmax? Could you please explain it a little more? Thank you.

3 Likes

[MLzz](#) 18 July 24, 2022, 9:05pm

In “Feedforward Neural Networks in Depth, Part1...”, equation (12), how do you proof this? It seems odd to me that summation of partial derivatives. Thanks in advance.

[anon57530071](#) 19 July 25, 2022, 3:52am

I think it's a simple multivariable calculus with a chain rule. Conditions given are;

$$\begin{aligned} u_i &= g_i(x_1, x_2, \dots, x_j, \dots, x_n) \\ y_k &= f_k(u_1, u_2, \dots, u_i, \dots, u_m) \end{aligned}$$

Then, a partial derivative of y_k with respect to x_j will be;

$$\begin{aligned} \frac{\partial y_k}{\partial x_j} &= \frac{\partial}{\partial x_j} f_k(u_1, u_2, \dots, u_m) \\ &= \frac{\partial}{\partial x_j} f_k(g_1(x_1, x_2, \dots, x_j, \dots, x_n), g_2(x_1, x_2, \dots, x_j, \dots, x_n), \dots, g_m(x_1, x_2, \dots, x_j, \dots, x_n)) \\ &= \frac{\partial f_k}{\partial g_1} \frac{\partial g_1}{\partial x_j} + \frac{\partial f_k}{\partial g_2} \frac{\partial g_2}{\partial x_j} + \dots + \frac{\partial f_k}{\partial g_m} \frac{\partial g_m}{\partial x_j} \\ &= \sum_i \frac{\partial f_k}{\partial g_i} \frac{\partial g_i}{\partial x_j} \end{aligned}$$

As $y_k = f_k()$, and $u_i = g_i()$, I think the equation (12) is correct.

3 Likes

[Venkat_Dhinakaran](#) 20 August 15, 2022, 3:34pm

If you differentiate equation (1) with respect to $w[l]$ you get that $a[l-1]$ in equation (13).

1 Like

[Manuel_Angelini](#) 21 August 31, 2022, 1:04pm

how can I download the articles in pdf format?

2 Likes

[Pablo_Daniel11](#) 22 September 1, 2022, 5:19pm

Thank you!

Is there any way to download the articles in pdf?

2 Likes

[Sami belattar](#) 23 September 30, 2022, 5:00pm

Thank You for your your detailed explanation

1 Like

[ChengHsien Li](#) 25 November 10, 2022, 8:48am

I have PhD degree in physics and I was eager to see a more generalized mathematical derivation of forward & backward propagation from the course videos. After gaining some basic understanding and intuition of what forward and backward propagation are for from the course videos, these notes are a great bonus to who are comfortable with multivariate calculus. I appreciate the amount of effort the course instructor put in to create these notes.

11 Likes

[Divyanshu Dimri](#) 26 January 24, 2023, 9:13pm

yep I thought so too, but if you observe it carefully its a same thing(As far as I can depict)

[Allaeddine Boudour](#) 27 March 7, 2023, 12:15pm

i didn't understand this point , because if j doesn't equal to p that mean the derivation equal 0.

where $\partial J / \partial \mathbf{z}_{:,i}^{[l]} \in \mathbb{R}^{n^{[l]}}$, $\partial J / \partial \mathbf{z}^{[l]} \in \mathbb{R}^{n^{[l]} \times m}$, $\partial J / \partial \mathbf{W}^{[l]} \in \mathbb{R}^{n^{[l]} \times n^{[l-1]}}$, and $\partial J / \partial \mathbf{b}^{[l]} \in \mathbb{R}^{n^{[l]}}$.

Looking back at (13) and (14), we see that the only unknown entity is $\partial J / \partial z_{j,i}^{[l]}$. By applying the chain rule once again, we get

$$\frac{\partial J}{\partial z_{j,i}^{[l]}} = \sum_p \frac{\partial J}{\partial a_{p,i}^{[l]}} \frac{\partial a_{p,i}^{[l]}}{\partial z_{j,i}^{[l]}}, \quad p \neq j \quad (17)$$

where $p = 1, \dots, n^{[l]}$.

Next, we present the vectorized version:

[rmwkwok](#) 28 March 9, 2023, 1:58am

Hi [@Allaeddine Boudour](#)

I only read a few lines of the first part and [this reply](#).

I think your understanding is correct for all hidden layers but there is exception in the output layer where softmax is used as the activation, because softmax uses z's from all nodes to compute each node's a, so in other words, each a will depend on all of the z's.

Cheers,
Raymond

1 Like

[Allaeddine Boudour](#) 29 March 10, 2023, 9:09pm

thank you for your explantation i appreciate it.

[rmwkwok](#) 30 March 11, 2023, 12:47am

You are welcome, [@Allaeddine Boudour](#) !

Cheers,
Raymond

[Yuechun](#) 31 May 30, 2023, 1:46am

This is amazing. This is awesome. Beyond my expectation!!! Thank you so much I like this this is so helpful!!! I love it!!! thank you!!!

[abdalla_ebrahim](#) 32 July 1, 2023, 8:53pm

Figure 2: A node in the previous layer.

In the future, we might want to write an implement from scratch in, for example, Python. To take advantage of the heavily optimized versions of vector and matrix operations that come bundled with libraries such as NumPy, we need to vectorize eq_z_scalar and eq_a_scalar .

To begin with, we vectorize the nodes:

```
\begin{align*} \begin{matrix} z_1, \dots, z_{j-1}, z_j, \dots, z_{n^l} \end{matrix} \\ \begin{matrix} w_1, \dots, w_{j-1}, w_j, \dots, w_{n^l} \end{matrix} \\ \begin{matrix} a_1, \dots, a_{j-1}, a_j, \dots, a_{n^l} \end{matrix} \\ \begin{matrix} b_1, \dots, b_{j-1}, b_j, \dots, b_{n^l} \end{matrix} \\ \begin{matrix} g_1, \dots, g_{j-1}, g_j, \dots, g_{n^l} \end{matrix} \end{matrix}
```

which we can write as

```
\begin{align*} \begin{matrix} z_1, \dots, z_{j-1}, z_j, \dots, z_{n^l} \end{matrix} &= \begin{matrix} w_1, \dots, w_{j-1}, w_j, \dots, w_{n^l} \end{matrix} \\ \begin{matrix} a_1, \dots, a_{j-1}, a_j, \dots, a_{n^l} \end{matrix} &= \begin{matrix} g_1, \dots, g_{j-1}, g_j, \dots, g_{n^l} \end{matrix} \end{matrix}
```

where $\begin{matrix} z_1, \dots, z_{j-1}, z_j, \dots, z_{n^l} \end{matrix}$ is the j th column of $\begin{matrix} z_1, \dots, z_{n^l} \end{matrix}$, and so on.

I am sorry, but it appears corrupted like in the picture.
does anyone have the same problem?

1 Like

[rmwkwok](#) 33 July 1, 2023, 9:49pm

I am not an expert at this but it might be related to the browser you're using. My Chrome works just fine. Would you try it on Chrome?

[abdalla_ebrahim](#) 34 July 1, 2023, 11:32pm

Yes, I am using Chrome latest version, and I tried on edge giving the same view.
also it gives the same view on my mobile.
thanks.

[rmwkwok](#) 35 July 2, 2023, 12:08am

I am sorry to hear that. I did a quick search with the keywords “chrome doesn’t render latex equation”, and found this:

there a way to test it for sure? – Tejas Dec 7, 2020 at 2:47

@hardmath Yes, I believe that is the javascript that needs to be accessible. I can open it too on a newer laptop, but the laptop I mentioned cannot reach that site. The Chrome version is 87.0.4280.88 on both the laptops. – Tejas Dec 7, 2020 at 2:49

5 @AsafKaragila I found that an antivirus was blocking some adware. Disabling it has enabled rendering of the Latex again! Thank you! – Tejas Dec 7, 2020 at 2:59

You see - antivirus might not be the source of your problem (this answer is also 2.5 years ago) but perhaps it can be lead for you to further investigate. You might also find other helpful hints if you google for more. It sounds going to be some work for you to figure out how to get it work.

Also, given that you have tried two devices and different browsers and none work. One thing in common (but you might know more) between all of your trials could be that they are on the same network that perhaps might share some common firewall settings that sounds to be quite related to antivirus thing.

Good luck to your investigations for yourself!

Cheers,
Raymond

[abdalla ebrahim](#) 36 July 2, 2023, 12:34am

I tried on mobile data, and turned the firewall off but still the same, anyway thank you very much for your response, and i will try to solve this issue.

Cheers,
Abdullah

[ramdhan ziane](#) 37 July 15, 2023, 2:24pm

Hello, I have a question about both the derivative of W and b ,
So by the definition of the chain rule

$$u_i = g_i(x_1, \dots, x_j, \dots, x_n), \quad (10)$$

$$y_k = f_k(u_1, \dots, u_i, \dots, u_m). \quad (11)$$

Note that x_j may affect $u_1, \dots, u_i, \dots, u_m$, and y_k may depend on $u_1, \dots, u_i, \dots, u_m$; thus,

$$\frac{\partial y_k}{\partial x_j} = \sum_i \frac{\partial y_k}{\partial u_i} \frac{\partial u_i}{\partial x_j}. \quad (12)$$

I find it understandable to get the result provided by the first article

$$\frac{\partial J}{\partial \mathbf{W}^{[l]}} = \sum_i \frac{\partial J}{\partial \mathbf{z}_{:,i}^{[l]}} \mathbf{a}_{:,i}^{[l-1]\top} = \frac{\partial J}{\partial \mathbf{Z}^{[l]}} \mathbf{A}^{[l-1]\top},$$

$$\frac{\partial J}{\partial \mathbf{b}^{[l]}} = \sum_i \frac{\partial J}{\partial \mathbf{z}_{:,i}^{[l]}} = \underbrace{\sum_{\text{axis}=1} \frac{\partial J}{\partial \mathbf{Z}^{[l]}}}_{\text{column vector}},$$

however in many cases including the formulas provided in the DLS course, the equations provided are

$$dW^{[l]} = \frac{\partial J}{\partial W^{[l]}} = \frac{1}{m} dZ^{[l]} A^{[l-1]\top}$$

$$db^{[l]} = \frac{\partial J}{\partial b^{[l]}} = \frac{1}{m} \sum_{i=1}^m dZ^{[l](i)}$$

$$dA^{[l-1]} = \frac{\partial \mathcal{L}}{\partial A^{[l-1]}} = W^{[l]\top} dZ^{[l]}$$

and I can't figure out why we divided by m in both dW and db , am I missing something or is it just convenience to divide by m given that we generated the result by summing the z over examples for each node.

Thanks in advance.

[saifkhanengr](#) 38 July 15, 2023, 2:28pm

ramdhan_ziane:

why we divided by m in both dW and db

Because there is m in the denominator of the cost function. And, dW and db are the derivatives of the cost function w.r.t. parameters, right?

1 Like

[ramdhan_ziane](#) 39 July 15, 2023, 3:08pm

Yes, you're right, the thing is when calculating the derivative of A at the final layer we used different formulas

$$\frac{\partial J}{\partial \mathbf{Z}^{[L]}} = \frac{1}{m} (\mathbf{A}^{[L]} - \mathbf{Y}).$$

$$dZ^{[L]} = A^{[L]} - Y$$

$$dW^{[L]} = \frac{1}{m} dZ^{[L]} A^{[L-1]T}$$

$$db^{[L]} = \frac{1}{m} np.sum(dZ^{[L]}, axis = 1, keepdims = True)$$

which explains why we do the dividing by m at each layer for W and b, so I think the more efficient way is using the first formula am I right?

[Alex Bleriot KASHIE](#) 40 July 27, 2023, 3:08pm

Wow, great job. Thank you Jonas

[FViloria](#) 41 July 28, 2023, 4:49pm

in the second post, i got lost trying to find dJ/dA in equation 4, dJ/dZ is expressed in function of dJ/dA but dJ/dA is not defined.

please explain

[saifkhanengr](#) 42 July 29, 2023, 4:59am

$\frac{dJ}{dZ}$ is equal to $\frac{dJ}{dA} \times \frac{dA}{dZ}$. So, in $\frac{dJ}{dZ}$, we covers the $\frac{dJ}{dA}$.

[JJaassoonn](#) 43 August 7, 2023, 10:24am

We will make heavy use of the chain rule in this section, and to understand better how it works, we first apply the chain rule to the following example:

$$u_i = g_i(x_1, \dots, x_j, \dots, x_n), \quad (10)$$

$$y_k = f_k(u_1, \dots, u_i, \dots, u_m). \quad (11)$$

Note that x_j may affect $u_1, \dots, u_i, \dots, u_m$, and y_k may depend on $u_1, \dots, u_i, \dots, u_m$; thus,

$$\frac{\partial y_k}{\partial x_j} = \sum_i \frac{\partial y_k}{\partial u_i} \frac{\partial u_i}{\partial x_j}. \quad (12)$$

$$\begin{aligned} \frac{\partial y_k}{\partial x_j} &= \frac{\partial}{\partial x_j} f_k(u_1, u_2, \dots, u_m) \\ &= \frac{\partial}{\partial x_j} f_k(g_1(x_1, x_2, \dots, x_j, \dots, x_n), g_2(x_1, x_2, \dots, x_j, \dots, x_n), \dots, g_m(x_1, x_2, \dots, x_j, \dots, x_n)) \\ &= \frac{\partial f_k}{\partial g_1} \frac{\partial g_1}{\partial x_j} + \frac{\partial f_k}{\partial g_2} \frac{\partial g_2}{\partial x_j} + \dots + \frac{\partial f_k}{\partial g_m} \frac{\partial g_m}{\partial x_j} \\ &= \sum_i \frac{\partial f_k}{\partial g_i} \frac{\partial g_i}{\partial x_j} \end{aligned}$$

Dear Mentors, May i know why should we add all terms together in this case?

Thank you.

[saifkhanengr](#) 44 August 7, 2023, 10:30am

Summation means summing all the terms. You highlighted the addition sign (+) in the second figure. All terms are adding starting from $i = 1$ to $i = m$.

[JJaassoonn](#) 45 August 7, 2023, 10:38am

Dear Mr Saif,

May i know why should we add all terms together in this case?

Thank you.

[saifkhanengr](#) 46 August 7, 2023, 11:25am

First, give us the background. What is the relationship of $u_{\{i\}}$ and $g_{\{i\}}(x_{\{1\}}, \dots, x_{\{j\}}, \dots, x_{\{n\}})$. Addition? And, $y_{\{k\}}$ & $f_{\{k\}}(u_{\{1\}}, \dots, u_{\{i\}}, \dots, u_{\{m\}})$.

[Amir Momenan](#) 47 August 18, 2023, 9:24am

Thank you for sharing, Your notes were very organized and useful. As an MSc student I understood most of it.

For calculating backward propagation equations, However, I prefer writing the chain rule wholly all the way from cost function to the target . For example, instead of writing 4 chain rules with the length of 1, I find it simpler to write a single line chain rule with the length of 4. It's basically the same thing but I find the second formation easier to understand.

Just trying to throw a suggestion out there. Thank you again for the amazing notes.

[Ernesto Cuartas Mora](#) 48 August 18, 2023, 11:00pm

U must calculate dZ/dW from eq (3), $Z[l] = W[l]A[l-1] + B[l]$. Where $dZ/dW = A[l-1]$

[Omar Mostafa1](#) 49 September 2, 2023, 10:21am

I am facing a problem with the articles.

the mathematical equations appear like this:

As a result, our definition of a node ends up introducing a fairly large number of notations:

$$\begin{aligned} z_{(j, i)}^{(l)} &= \sum_k w_{(j, k)}^{(l)} a_{(k, i)}^{(l-1)} + b_{(j, i)}^{(l)}, \quad \text{\label{eq:z_scalar} \ \ a_{(j, i)}^{(l)} = g_j^{(l)}(z_{(1, i)}^{(l)}, \dots, z_{(j, i)}^{(l)}, \dots, z_{(n^{(l)}, i)}^{(l)})} \end{aligned} \quad \text{\label{eq:a_scalar} }$$

Does the node definition look intimidating to you at first glance? Do not worry. Hopefully, it will make more sense once we have explained the notations, which we shall do next:

Entity	Description
$\backslash(l)$	The current layer $\backslash(l = 1, \dots, L)$, where $\backslash(L)$ is the number of layers that have weights and biases. We use $\backslash(l = 0)$ and $\backslash(l = L)$ to denote the input and output layers.
$\backslash(n^{(l)})$	The number of nodes in the current layer.

[aalu](#) 50 September 15, 2023, 10:15pm

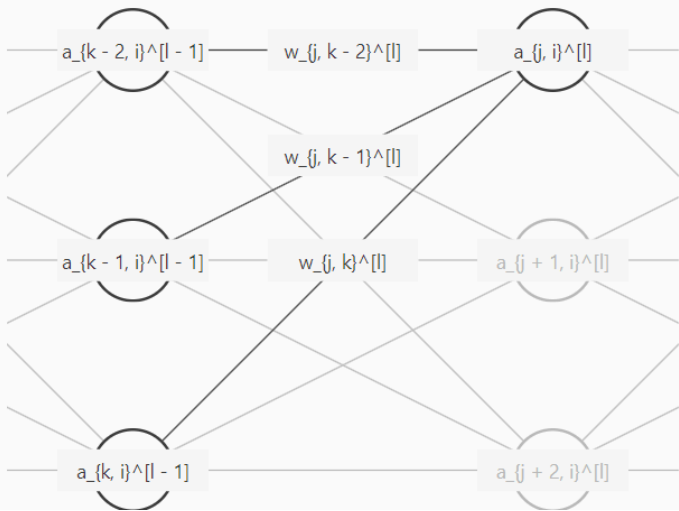
Hi, could you clarify the notation for f in (9) in part 1? I was expecting f to be $f: \mathbb{R}^{n^L \times m} \times \mathbb{R}^{n^L \times m} \rightarrow \mathbb{R}$, but it is $f: \mathbb{R}^{2n^L} \rightarrow \mathbb{R}$

[Bob Levy](#) 51 September 22, 2023, 11:19am

Any idea why the MathML embedded in the neural net SVGs here would fail to render? I get the same issue in Chrome, Firefox and MS Edge,

$a_{j,i}^{[l]}$	An activation in the current layer.
$a_{k,i}^{[l-1]}$	An activation in the previous layer.
$g_j^{[l]}$	An activation function $g_j^{[l]}: \mathbb{R}^{n^{[l]}} \rightarrow \mathbb{R}$ used in the current layer.

To put it concisely, a node in the current layer depends on every node in the previous layer, and the following visualization can help us see that more clearly:



[peppy](#) 52 October 13, 2023, 12:10am

I was confused by this when I first saw this as well. I found this article that might be helpful:



14.5: The Chain Rule for Multivariable Functions

In single-variable calculus, we found that one of the most useful differentiation rules is the chain rule, which allows us to find the derivative of the composition of two functions. The same thing ...

(see section titled " The Generalized Chain Rule")

1 Like

[kiran kumar Savvana](#) 53 October 13, 2023, 1:28am

Thanks for wonderful work.

[rmwkwok](#) 54 October 13, 2023, 1:38am

Thank you for sharing, [@peppy](#).

[manaviitd](#) 55 November 8, 2023, 5:59am

Thanks for the derivation. I initially thought after reading part 1 article that equation #17 was an overkill given that for any layer l , any unit j in that later, any training example i , the activation function $a_j^{[l]}(i)$ would be a function of *only* $z_j^{[l]}(i)$ and not any other $a_p^{[l]}(i)$. However after reading part 2 article and in particular the softmax function, I understood your motivation behind equation 17. Thanks much!

1 Like

[Shivang_Shekhar1](#) 56 March 19, 2024, 12:30pm

Coolest thing I have seen today

1 Like

[Francis60](#) 57 April 5, 2024, 4:21pm

Hi,

I asked myself the same question and even send a personal e-mail to [@jonaslalin](#) about it before i read this conversation.

Thank for the explantation [@rmwkwok](#) .

Apologies for cluttering his mailbox.

Francis

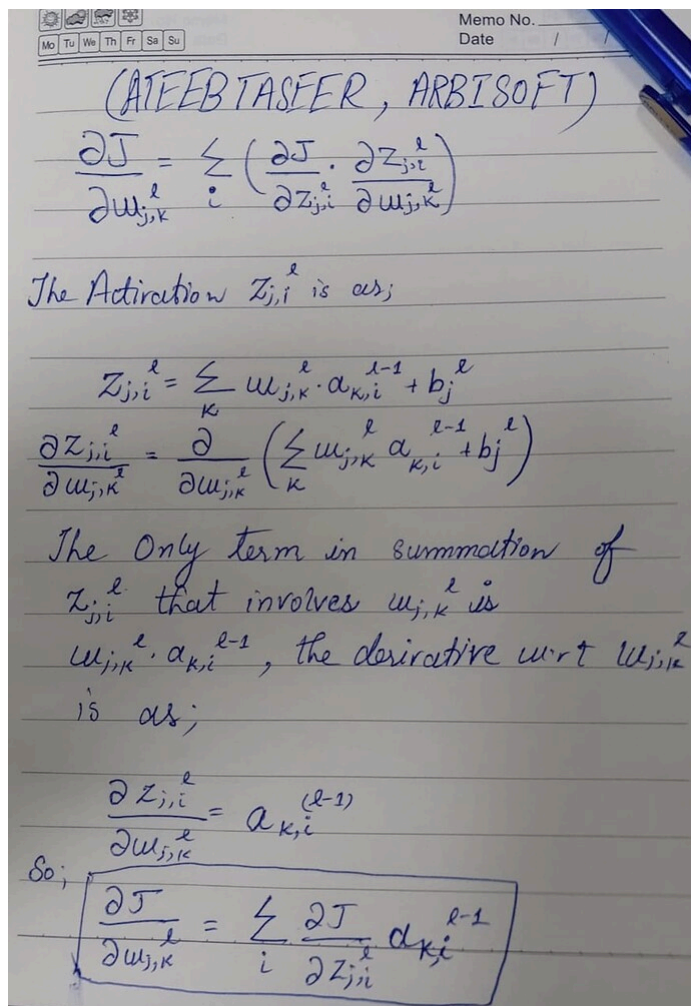
1 Like

[Francis60](#) 58 April 5, 2024, 5:04pm

No, but you can save as html. there are tools to parse html to pdf but it is touchy.

[ateebai](#) 60 April 16, 2024, 10:16am

Hope it Helps:



1 Like

[nitesh_garg](#) 61 June 25, 2024, 9:44am

Hi, I had a doubt regarding the cost function of multi-class classification. Doesn't this J give higher priority to classes having higher value as it is multiplied by y and have $\log(a[L])$? Does it work only because of soft-max activation function as if we use any other function, its derivative with respect to $Z[L]$ may not come out to be $1/m (A[L] - Y)$?

[Prashant360](#) 62 August 12, 2024, 1:17pm

We would also like to establish two additional notations:

$$\mathbf{A}^{[0]} = \mathbf{X}, \quad (7)$$

$$\mathbf{A}^{[L]} = \hat{\mathbf{Y}}, \quad (8)$$

where $\mathbf{X} \in \mathbb{R}^{n^{[0]} \times m}$ denotes the inputs and $\hat{\mathbf{Y}} \in \mathbb{R}^{n^{[L]} \times m}$ denotes the predictions/outputs.

Finally, we are ready to define the cost function:

$$J = f(\hat{\mathbf{Y}}, \mathbf{Y}) = f(\mathbf{A}^{[L]}, \mathbf{Y}), \quad (9)$$

where $\mathbf{Y} \in \mathbb{R}^{n^{[L]} \times m}$ denotes the targets and $f: \mathbb{R}^{2n^{[L]}} \rightarrow \mathbb{R}$ can be tailored to our needs.

We are done with forward propagation! Next up: backward propagation, also known as backpropagation, which involves computing the gradient of the cost function with respect to the weights and biases.

[paulinpaloalto](#) 63 August 12, 2024, 2:55pm

The author Jonas Slalin is the real expert here, but I'm not sure whether he is still answering questions here. If you don't get an answer from him, you could check his website (where this material is posted) and see if he has a link there for posting questions.

Are you asking why the domain of function f is $\mathbb{R}^{2n^{[L]}}$ as opposed to $\mathbb{R}^{n^{[L]}}$ or are you asking what $n^{[L]}$ means there?

If the latter, it is the number of neurons in the output layer of the network.

For the former question, my reading would be that the point is that both $\mathbf{A}^{[L]}$ and $\mathbf{Y}$ have dimension $n^{[L]} \times m$. So I would have thought the m would need to figure in there. I haven't read the rest of Jonas's definitions in a couple of years, but it looks like he is using the same convention that Prof Andrew Ng does that J is the scalar cost function which is the average of the vector loss values L across the samples. So I would have thought that you needed to incorporate the m into the dimension of the domain, but maybe since taking the average is basically trivial we can ignore that. Then we'd be down to just the discretionary part of the function being the vector loss which has $2n^{[L]}$ inputs.

But all this is just notation in any case. When you define the actual functions in question, this should all be clear and will just "come out in the wash".

1 Like

[ai_curious](#) 64 August 12, 2024, 4:20pm

paulinpaloalto:

I'm not sure whether he is still answering questions here.

His profile indicates last post date was May 2022

1 Like

[scossar](#) 65 November 13, 2024, 7:15am

This is great! I especially liked the subscripts j and k as a way of keeping track of what's going on.

The suggested 10 minutes reading time on the course website is a little misleading. I have no background in calculus and I gave it a day - with ChatGPT as a tutor. I think we got somewhere, but I'm going to have to come back to it a few times.

1 Like

[mehdiardavan](#) 66 November 22, 2024, 1:30pm

Hi Jonas. Thanks so much for these three articles "Feedforward Neural Networks in Depth". They are really great except the pixelated quality of the equations. I assume you wrote these in LaTeX and converted to HTML? Do you have the original .tex files or the PDF files to share?

[Li Tang](#) 67 January 25, 2025, 4:38am

Thanks for the resource for the further study, I have a question in part 2

Next, we compute the partial derivatives:

$$\begin{aligned}
 \frac{\partial a_{j,i}^{[l]}}{\partial u_7} &= 1, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_6} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_7} \frac{\partial u_7}{\partial u_6} = u_4 = \exp(z_{j,i}^{[l]}) - \exp(-z_{j,i}^{[l]}), \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_5} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_6} \frac{\partial u_6}{\partial u_5} = -u_4 \frac{1}{u_5^2} = -\frac{\exp(z_{j,i}^{[l]}) - \exp(-z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_4} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_7} \frac{\partial u_7}{\partial u_4} = u_6 = \frac{1}{\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]})}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_3} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_4} \frac{\partial u_4}{\partial u_3} + \frac{\partial a_{j,i}^{[l]}}{\partial u_5} \frac{\partial u_5}{\partial u_3} \\
 &= -u_6 - u_4 \frac{1}{u_5^2} \\
 &= -\frac{1}{\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]})} - \frac{\exp(z_{j,i}^{[l]}) - \exp(-z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2} \\
 &= -\frac{2 \exp(z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_2} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_4} \frac{\partial u_4}{\partial u_2} + \frac{\partial a_{j,i}^{[l]}}{\partial u_5} \frac{\partial u_5}{\partial u_2} \\
 &= u_6 - u_4 \frac{1}{u_5^2} \\
 &= \frac{1}{\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]})} - \frac{\exp(z_{j,i}^{[l]}) - \exp(-z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2} \\
 &= \frac{2 \exp(-z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2}, \\
 \frac{\partial a_{j,i}^{[l]}}{\partial u_1} &= \frac{\partial a_{j,i}^{[l]}}{\partial u_3} \frac{\partial u_3}{\partial u_1} \\
 &= \left(-u_6 - u_4 \frac{1}{u_5^2}\right) \exp(u_1) \\
 &= -\frac{2 \exp(z_{j,i}^{[l]}) \exp(-z_{j,i}^{[l]})}{(\exp(z_{j,i}^{[l]}) + \exp(-z_{j,i}^{[l]}))^2}
 \end{aligned}$$

Why the partial derivative for u3 equal to this?

[madhusanka.slc](#) 68 February 1, 2025, 11:39am

A well-explained and detailed breakdown of forward and backward propagation! Appreciate you sharing this with us.

[rmwkwok](#) 69 February 1, 2025, 12:15pm

Hello, [@Li Tang](#) ,

You might want to check out a topic called “total derivative” first, and with that in mind, see if the underlined equation is complete and correct.

Cheers,
Raymond

[fares26597](#) 70 July 13, 2025, 9:36am

Late to the party, but if I understand what you're saying correctly (still at Article 1), I think it would perhaps make things less confusing if, in Figures 1 and 2, multiplying by the W weights just leads to Z nodes instead of A nodes.

And then if you'd like you can add an extra layer to the right where we have A nodes, where each A node takes all Z nodes.

I know it's unorthodox to separate Z s and A s into different layers, but as it is currently, there isn't an a clear implication that each A node is made up of all the Z s in its same layer.

I hope my description isn't too confusing, and there could perhaps be better solutions, maybe a simple note beneath the figures, I don't know. But all in all, the content is great, thank you!

Edit: when you mention the NumPy documentation, are you just referring to the regular one on the NumPy website or is there like a PDF made for this course?

[paulinpaloalto](#) 71 July 13, 2025, 2:25pm

There is no special PDF of the Numpy documentation for these courses, so you need to consult the numpy website. One other important thing to note is that these courses typically use the versions of all software packages (python, numpy, TF, all libraries) that were current at the time the courses were last updated, which is April 2021 for most of the courses in DLS. The various software packages continue to evolve, so the documentation on the numpy website (for example) will show the current state of play and thus may have features that were not available when the courses were published.

Also note that checking Jonas's forum activity, he last posted anything in 2022, so you probably will not get any answers from him at least here in this context.

1 Like

[fares26597](#) 72 July 13, 2025, 3:09pm

I see, thanks for letting me know!

[sage2003](#) 73 September 13, 2025, 9:48pm

Thank you for writing the blog, Jonas!

After watching the course videos, I was looking for an in-depth derivation and explanation of the forward and backward propagation, and this provides exactly that without wandering around much. I'm sure this will help many other students like me. Highly appreciate the effort put into writing this and providing the information for free!

1 Like

[YuJingyao](#) 74 September 20, 2025, 8:20am

splendid explanation! Thanks for your clarification. I couldn't be more clear about those basic concepts in neural networks.

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Typo in back prop formula (week3 and week 4) coursera-platform	7	December 10, 2021
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